PRODUCT USER MANUAL

Multi Observation Global Ocean Sea Surface Salinity Optimal Interpolation Product

MULTIOBS_GLO_PHY_SSS_L4_MY_015_015

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RECORD TABLE

Issue	Date	§	Description of Change	Author	Validated By
1.0	2022/06/15	All	Creation of the document	S. Tarot	Copernicus Marine product management
2.0	2024/01/29	All	Extension of the series to SMOS+SMAP data	S. Tarot	Copernicus Marine product management

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GLOSSARY AND ABBREVIATIONS

ATBD	Algorithm Theorical Basis Document
CATDS	Centre Aval de Traitement des Données du Satellite SMOS (Processing centre for high-level data from the SMOS satellite)
CF	Climate Forecast (convention for NetCDF)
FTP	File Transfer Protocol
GIS	Geographic Information System
ISAS	In-Situ Analysis Software
LOPS	Laboratoire d'Océanographie Physique et Spatiale (Laboratory for Ocean Physics and Satellite remote sensing)
MFC	Monitoring and Forecasting Centre
NetCDF	Network Common Data Form
PUM	Product User Manual
SMAP	Soil Moisture Active and Passive (name of a satellite)
SMOS	Soil Moisture – Ocean Salinity (name of a satellite)

I INTRODUCTION

I.1 Summary

This document is the Product User Manual (PUM) describing the external product MULTIOBS_GLO_PHY_SSS_L4_MY_015_015, developed by the CATDS Production Unit (MULTIOBS-CATDS-BREST-FR). This product provides Level-4 (L4) maps of Sea Surface Salinity obtained by Optimal Interpolation of data from the Soil Moisture and Ocean Salinity (SMOS) satellite, the Soil Moisture Active and Passive (SMAP) satellite, and In-Situ observations.

The CATDS (Centre Aval de Traitement des Données SMOS) is the French ground-segment facility in charge of the generation, calibration, archiving and dissemination of the SMOS level 3 and level 4 science products. It processes Sea Surface Salinity (SSS, also named Ocean Salinity, OS) and Soil Moisture. CATDS is also providing services to users and scientists.

The product MULTIOBS_GLO_PHY_SSS_L4_MY_015_015 is a reformatting and a simplified version of the CATDS L4 product called “SMOS-OI”. This product is obtained using optimal interpolation (OI) algorithm, that combine, ISAS in situ SSS OI analyses to reduce large scale and temporal variable bias, SMOS satellite image, SMAP satellite image, and satellite SST information. The ISAS in situ SSS OI analyses used are the Copernicus Marine Service products INSITU_GLO_PHY_TS_OA_NRT_013_002 for NRT, and the ISAS reanalysis from LOPS (Laboratoire d'Océanographie Physique et Spatiale) for MY.

The manual covers: how the product was created, its content, and which data services are available to access it.

Information on operational issues on products and services can be found on our User Notification Service. If you have any questions, please contact us.

I.2 History of changes

Date	**Description of changes and impacted product**
November 2022	First version of the product MULTIOBS_GLO_PHY_SSS_L4_MY_015_015
June 2024	Extension of the series to SMOS+SMAP data

II DESCRIPTION OF THE PRODUCT SPECIFICATION

II.1 General Information

Product Lines	MULTIOBS_GLO_PHY_SSS_L3_MYNRT_015_015
Geographical coverage	Global: 83.517°S-83.517°N, 179.87°W-179.87°E
Variables	Sea surface salinity Sea surface temparature Spiciness Absolute salinity Conservative temperature Density Thermal expansion coefficient Haline contraction coefficient
Product Type	Satellite observation
Available time series	2010/05/31 – up to Month-5
Temporal resolution	Weekly
Target delivery time	See II.6 Processing information
Delivery mechanism	Copernicus Marine Service Information Service
Horizontal resolution	~25km at 30° latitude on the EASE 2.0 cylindrical grid
Number of vertical levels	N/A (sea surface data)
Format	NetCDF 4.0 CF1.8

Table 1: MULTIOBS_GLO_PHY_SSS_L4_MY_015_015 product specification

II.2 Details of datasets

MULTIOBS_GLO_PHY_SSS_L4_MY_015_015
Dataset: **cmems_obs-mob_glo_phy-sss_my_multi-oi_P1W** **Variables name in the NetCDF file and Unit: Long_name & Standard_name**
sss [0.001] Corrected Practical Sea Surface Salinity (pss-78) sea_surface_salinity
sss_isas [0.001] Reference Practical SSS from ISAS in situ gridded product (pss-78) sea_surface_salinity
sss_corr_smos [0.001] Practical Sea Surface Salinity Correction of SMOS (pss-78) -
sss_corr_smap [0.001] – *for SMAP availability period* Practical Sea Surface Salinity Correction of SMAP (pss-78) -
sst [Celsius] Sea Surface Temperature sea_surface_temperature
spiciness0 [kg/m^3] Spiciness at p = 0 dbar (75-term equation) computed from TEOS-10 MATLAB Toolbox -
SA [g/kg] Absolute Salinity from Practical Salinity computed from TEOS-10 MATLAB Toolbox sea_surface_salinity
CT [Celsius] Conservative Temperature from in-situ temperature computed from TEOS-10 MATLAB Toolbox sea_water_conservative_temperature
rho [kg/m^3] In-situ density computed from TEOS-10 MATLAB Toolbox sea_surface_density
alpha [1/K] Thermal expansion coefficient with respect to Conservative Temperature computed from TEOS-10 MATLAB Toolbox -
beta [1000] Haline contraction coefficient at constant Conservative Temperature computed from TEOS-10 MATLAB Toolbox -

Table 2: list of the datasets and variable names and unit for the MULTIOBS_GLO_PHY_SSS_L4_MY_015_015 product

II.3 Production System Description

The production system is described in the Algorithm Theorical Basis Document (ATBD), available at https://www.catds.fr/Resources/Documentation : ATBD L3OS/L4OS – description of the algorithms of the L3.L4 Ocean Salinity products.

II.4 Grid

The product MULTIOBS_GLO_PHY_SSS_L4_MY_015_015 is based on the global cylindrical EASE-Grid 2.0 (Equal-Area Scalable Earth Grids – https://nsidc.org/data/user-resources/help-center/guide-ease-grids).

With this grid, the longitude steps are regular (~0.259°) but the latitude steps are not (~0.196° at equator, ~1.534° near poles). With the grid information provided in the files, most of the GIS tools are able to handle this grid. However, there are still some exceptions.

II.5 Vertical Levels

The vertical level is not indicated in the files. All the data are surface data.

II.6 Processing information

II.6.1 Processing steps

There are 3 steps of processing for each week of observation (D):

- The first processing step is not used for this product.
- The second processing step is performed once a month when the ISAS analysis is available.
- The third processing step correspond to the CATDS reprocessing campaigns. Such data replaces the data from the previous steps.

The processing step of a particular file is indicated in the global attribute processing step, with the values :

- “2 – operational processing”
- “3 – reprocessing”

It is also indicated in the filename, when downloaded through the Copernicus Marine Service FTP Service (See IV.2 Nomenclature of files when downloaded through the Copernicus Marine Service FTP Service).

II.6.2 Satellite unavailability

Due to satellite missions hazards, the data of one of the satellites is sometimes missing for a certain period of time (for example, in July 2019, SMAP was in safe-mode). In such case, the unavailability is indicated in the comment attribute.

For example:

```
:comment = "config_126 : SMOS_RE06_L3Q-SMAPV5.3_70km-9d-gauss-varJLV-OI1-V1-500-1-0-OI2-25-1-0.2-medSTD; source_3 (SMAP data) missing"
```

III DOWNLOAD A PRODUCT

After registration, you will be able to download our data. To assist you, our HelpCenter is available, and more specifically its section about download.

Information on operational issues on products and services can be found on our User Notification Service. If you have any questions, please contact us.

IV FILES NOMENCLATURE

IV.1 Nomenclature of files when downloaded through the Copernicus Marine Service Web Portal Subsetter Service

Files nomenclature when downloaded through the Copernicus Marine Service Web Portal Subsetter is based on product dataset name and a numerical reference related to the request date on the MIS.

The scheme is: **datasetname_nnnnnnnnnnnnn.nc**

where:

- **datasetname** is cmems_obs-mob_glo_phy-sss_my_multi-oi_P1W
- **nnnnnn**: 6 digit integer corresponding to the current time (download time) in millisec-onds since January 1, 1970 midnight UTC.
- **.nc**: standard NetCDF filename extension.

Example: cmems_obs-mob_glo_phy-sss_myt_multi-i_P1W_1318943720287.nc

IV.2 Nomenclature of files when downloaded through the Copernicus Marine Service FTP Service

Files nomenclature when downloaded through the Copernicus Marine Service FTP Service is based as follows:

- ***YYYY*/CATDS_CSF40n_*YYYYMMDD*T000000_*YYMMDD*T235959_*VVV*_*S*.nc**

Where:

- **YYYY:** the year of observation
- **CSF40n**: the CATDS code of the product, with
  - n=1 for SMOS only period (2010 – 03/2015)
  - n=2 for SMOS+SMAP period (04/2015 – ongoing)
- **YYYYMMDD**: the date of observation
- **VVV**: the version of the processor
- **S**: is the processing step (See II.6 Processing information)
- **.nc**: standard NetCDF filename extension

IV.3 File size

Each file is about 71 Mb.

V FILE FORMAT

V.1 NetCDF

The products are stored using the NetCDF format.

The products are stored using the NetCDF format. To know more about the NetCDF format, please follow this link: [What is the format of Copernicus Marine products ? NetCDF]

V.2 Structure and semantic of NetCDF maps files

An example header of a MULTIOBS_GLO_PHY_SSS_L4_MY_015_015 file (generated using ncdump) is given below.

```
netcdf CATDS_CSF4O2_20210809T000000_20210815T235959_345_3 {
dimensions:
  lat = 584 ;
  lon = 1388 ;
  time = UNLIMITED ;
variables:
  double lat(lat) ;
    lat:long_name = "latitude" ;
    lat:units = "degrees_north" ;
    lat:axis = "Y" ;
    lat:standard_name = "latitude" ;
  double lon(lon) ;
    lon:long_name = "longitude" ;
    lon:units = "degrees_east" ;
    lat:axis = "X" ;
    lon:standard_name = "longitude" ;
  double time(time) ;
    time:long_name = "mean time of coverage period" ;
    time:units = "days since 2000-01-01T00:00:00Z" ;
    time:standard_name = "time" ;
    time:axis = "T" ;
    time:calendar = "standard" ;
  double sss(time, lat, lon) ;
    sss:long_name = "Corrected Practical Sea Surface Salinity (pss-78)" ;
    sss:standard_name = "sea_surface_salinity" ;
    sss:units = "0.001" ;
  double sss_isas(time, lat, lon) ;
    sss_isas:long_name = "Reference Practical SSS from ISAS in situ gridded product (pss-78)" ;
    sss_isas:standard_name = "sea_surface_salinity" ;
    sss_isas:units = "0.001" ;
  double sss_corr_smos(time, lat, lon) ;
    sss_corr_smos:long_name = "Practical Sea Surface Salinity Correction of SMOS (pss-78)" ;
    sss_corr_smos:units = "0.001" ;
  double sss_corr_smap(time, lat, lon) ;
    sss_corr_smos:long_name = "Practical Sea Surface Salinity Correction of SMAP (pss-78)" ;
    sss_corr_smos:units = "0.001" ;
  double sst(time, lat, lon) ;
    sst:long_name = "Sea Surface Temperature" ;
    sst:standard_name = "sea_surface_temperature" ;
    sst:units = "Celsius" ;
  double spiciness0(time, lat, lon) ;
    spiciness0:long_name = "Spiciness at p = 0 dbar (75-term equation) computed from TEOS-10
MATLAB Toolbox" ;
    spiciness0:units = "kg/m^3" ;
  double SA(time, lat, lon) ;
```

```
    SA:long_name = "Absolute Salinity from Practical Salinity computed from TEOS-10 MATLAB
Toolbox" ;
    SA:standard_name = "sea_surface_salinity" ;
    SA:units = "g/kg" ;
  double CT(time, lat, lon) ;
    CT:long_name = "Conservative Temperature from in-situ temperature computed from TEOS-10 MATLAB
Toolbox" ;
    CT:standard_name = "sea_water_conservative_temperature" ;
    CT:units = "Celsius" ;
  double rho(time, lat, lon) ;
    rho:long_name = "In-situ density computed from TEOS-10 MATLAB Toolbox" ;
    rho:standard_name = "sea_surface_density" ;
    rho:units = "kg/m^3" ;
  double alpha(time, lat, lon) ;
    alpha:long_name = "Thermal expansion coefficient with respect to Conservative Temperature
computed from TEOS-10 MATLAB Toolbox" ;
    alpha:units = "1/K" ;
  double beta(time, lat, lon) ;
    beta:long_name = "Haline contraction coefficient at constant Conservative Temperature computed
from TEOS-10 MATLAB Toolbox" ;
    beta:units = "kg/g" ;
  double ice_mask(time, lat, lon) ;
    ice_mask:long_name = "Sea Ice Fraction (>=15 %) from multi-sensor L4 foundation SST product
from Remote Sensing Systems" ;
    ice_mask:_FillValue = 0. ;
    ice_mask:mask_data = 1. ;

// global attributes:
    :title = "SSS SMOS/SMAP L4 OI - LOPS-v2023" ;
    :Conventions = "CF-1.8" ;
    :netcdf_version_id = "3.6.2" ;
    :creation_date = "08-Jan-2024 12:47:17" ;
    :product_version = "1.0" ;
    :institution = "SMOS CATDS & LOPS (www.catds.fr)" ;
    :source_1 = "Coriolis ISAS NRT" ;
    :references_1 = "http://marine.copernicus.eu/" ;
    :source_2 = "CATDS-PDC L3OS L3Q mixed - Debiased average 10 days salinity field product" ;
    :references_2 = "http://dx.doi.org/10.12770/0f02fc28-cb86-4c44-89f3-ee7df6177e7b" ;
    :source_3 = "SMAP SSS - V05.3 Validated Release" ;
    :references_3 = "http://www.remss.com/missions/smap/salinity/" ;
    :source_4 = "Remote Sensing Systems - AMSR-2 - SST" ;
    :references_4 = "http://www.remss.com/missions/amsr/" ;
    :source_5 = "Remote Sensing Systems - REMSS-L4_GHRSST-SSTfnd-MW_OI-GLOB-v02.0-fv05.1 - Sea
Ice Fraction" ;
    :references_5 = "http://www.remss.com/measurements/sea-surface-temperature/oisst-
description/" ;
    :computation_method = "TEOS-10" ;
    :references = "IOC, SCOR and IAPSO, 2010: The international thermodynamic equation
of seawater - 2010: Calculation and use of thermodynamic properties. Intergovernmental
Oceanographic Commission, Manuals and Guides
    :comment = "config_126 : SMOS_RE06_L3Q-SMAPV5.3_70km-9d-gauss-varJLV-OI1-V1-500-1-0-OI2-25-
1-0.2-medSTD" ;
    :ease_projection = "cylindrical" ;
    :ease_resolution = 25. ;
    :ease_origin_lat = -83.51714 ;
    :ease_origin_lon = -179.8703 ;
    :ease_global = "yes" ;
    :grip_mapping = "projection" ;
    :datum = "+ellps = WGS84" ;
    :srid = "EPSG:6933" ;
    :proj4text = "+proj=cea +lon_0=0 +lat_ts=30 +x_0=0 +y_0=0 +ellps=WGS84 +units=m +no_defs" ;
    :DOI = "10.12770/639a918e-0711-4eff-965d-15a728acb27d" ;
    :DOI_link = "https://doi.org/10.12770/639a918e-0711-4eff-965d-15a728acb27d" ;
    :citation = "CATDS (2024). CATDS-PDC L4OS SMOS/SMAP-OI - Weekly optimal interpolation
salinity field product from SMOS and SMAP satellites and ISAS. CATDS (CNES, IFREMER, LOPS).
http://dx.doi.org/10.12770/639a918e-0711-4eff-965d-15a728acb27d"
    :publication = "In addition to properly cite this dataset, it would be appreciated that the
following work(s) be cited too, when using this dataset in a publication: Kolodziejczyk
Nicolas, Hamon Michel, Boutin Jacqueline, Vergely Jean-Luc, Reverdin Gilles, Supply Alexandre,
Reul Nicolas (2021). Objective analysis of SMOS and SMAP Sea Surface Salinity to reduce large
scale and time dependent biases from low to high latitudes. Journal Of Atmospheric And Oceanic
```

Technology, 38(3), 405-421. Publisher\'s official version : https://doi.org/10.1175/JTECH-D-20-0093.1 , Open Access version : https://archimer.ifremer.fr/doc/00665/77702/" ;
}

VI REFERENCES

Kolodziejczyk Nicolas, Hamon Michel, Boutin Jacqueline, Vergely Jean-Luc, Reverdin Gilles, Supply Alexandre, Reul Nicolas (2021). Objective analysis of SMOS and SMAP Sea Surface Salinity to reduce large scale and time dependent biases from low to high latitudes. Journal Of Atmospheric And Oceanic Technology, 38(3), 405-421. Publisher's official version : https://doi.org/10.1175/JTECH-D-20-0093.1, Open Access version : https://archimer.ifremer.fr/doc/00665/77702/